

General information on software engineering

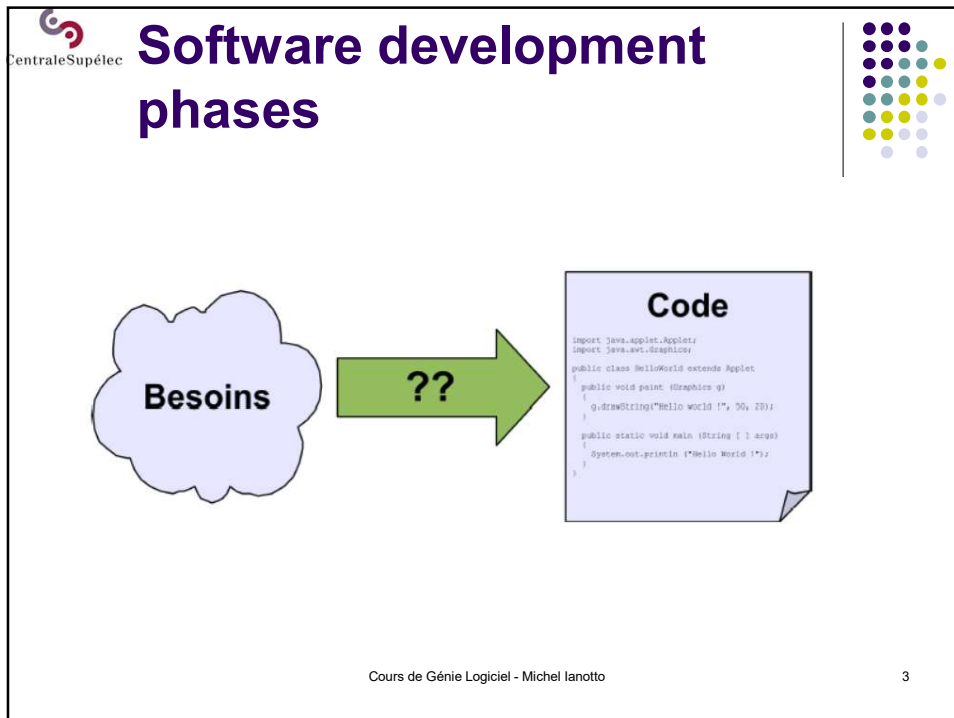
1. Definition of Software Engineering
2. Software development phases
3. Software development lifecycle
4. Software Development Approaches
5. Software



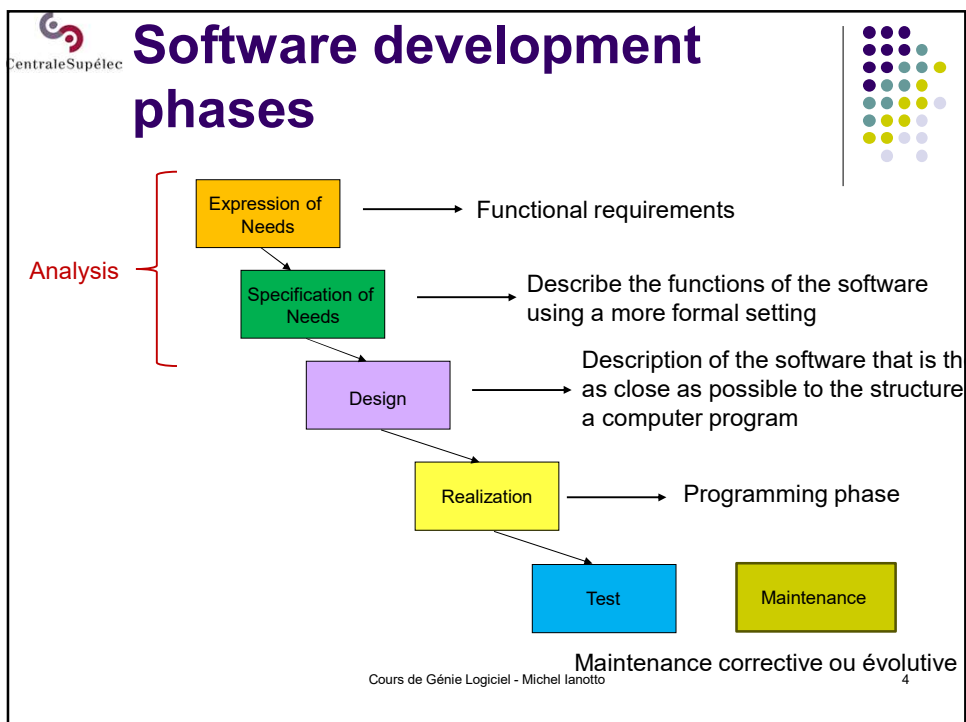
Definition of Software Engineering

- **Software engineering**: a discipline that concerns the development and evolution of software.
 - A discipline that dates back to the 1970s
- **Goal of software engineering**: to propose methods, techniques and tools that make it possible to specify, develop and evolve software:
 - meeting the needs of users,
 - reliable,
 - within the prescribed timeframe.





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Software development cycles

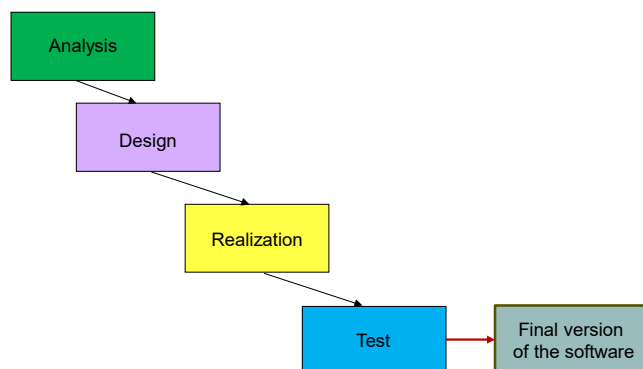


- The purpose of the software development cycle is to explain how the different stages of software development are linked.
- Major development cycles
 - Sequential cycle
 - Iterative cycle
 - V-cycle

Software development cycles



- Sequential development cycle



Software development cycles

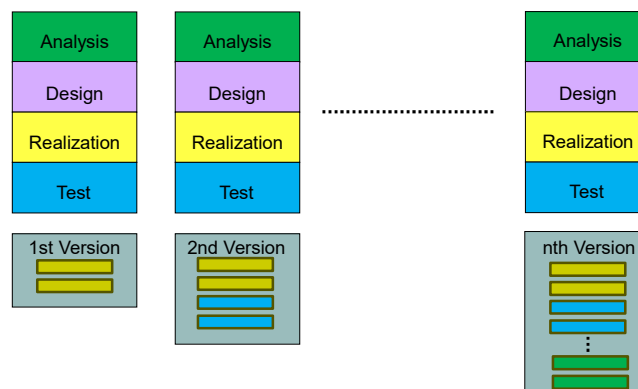


- Sequential development Cycle:
 - User needs **clearly identified** at the beginning of the project
 - Needs do not change during the project
 - Small projects
- Disadvantages :
 - **Rigid** design
 - Errors can be seen quite late

Software development cycles



- **Iterative** development cycle





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Software development cycles



- Iterative and incremental development cycle:
 - Big projects
 - User needs are **poorly identified** at the start of the project
 - User needs are likely **to evolve** during the course of the project
- Disadvantages :
 - Fine-tune increment sizes

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Software development cycles



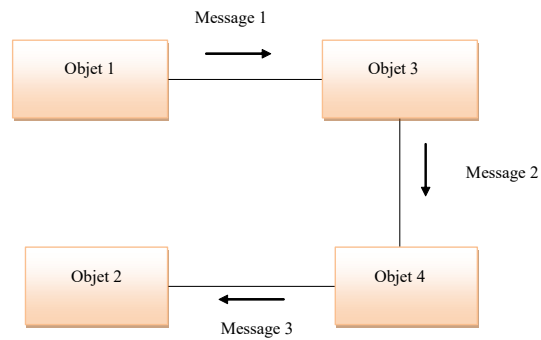
- **V-Cycle**
 - Evolution of the sequential development cycle
 - Testing activities are prepared in parallel with development activities

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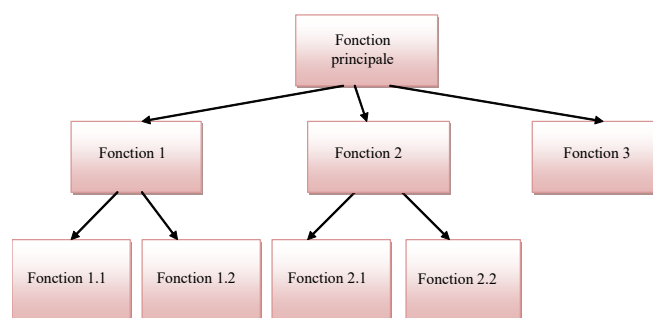
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Object approach



- Software Functions
 - Objects + messages

Approche fonctionnelle



- Hierarchy of functions
- Data structure

Software



- **Software** is a set of programs that allows a computer or computer system to perform a **specific task** or **function**.
 - word processor or spreadsheet,
 - application that manages the number of points on the driving license,
 - antivirus,
 - C++ code compiler.

Software



- Not just **the executable code** of a program
- Design, development, and maintenance **documentation**
- Installation and usage **documentation**
- Documented **source code**